

Whole-cell patch-clamp measurements of spermatozoa reveal an alkaline-activated Ca^{2+} channel

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In mammals, sperm cells become motile during ejaculation and swim up the female reproductive tract. Before fertilization and to overcome various barriers, their motility must be hyperactivated, a motion that is characterized by vigorous asymmetric tail beating¹. Hyperactivation requires an increase in calcium in the flagella, a process that probably involves plasmalemmal ion channels^{2–8}. Numerous attempts in the past two decades to understand sperm cell channels have been frustrated by the difficulty of measuring spermatozoan transmembrane ion currents^{2,3,9–16}. Here, by using a simple approach to patch-clamp spermatozoa and to characterize whole-spermatozoan currents, we describe a constitutively active flagellar calcium channel that is strongly potentiated by intracellular alkalinization. This current is not present in spermatozoa lacking the sperm-specific putative ion channel protein, CatSper1. This plasma membrane protein of the six transmembrane-spanning ion channel superfamily is specifically localized to the principal piece of the sperm tail and is required for sperm cell hyperactivation and male fertility^{4,5}. Our results identify CatSper1 as a component of the key flagellar calcium channel, and suggest that intracellular alkalinization potentiates CatSper current to increase intraflagellar calcium and induce sperm hyperactivation.

CatSper1, and three homologous proteins (CatSper2–CatSper4), comprise a family of sperm-specific proteins^{4,7,17}. In analogy to other members of the six transmembrane (6TM) channel family, CatSper proteins seem to be subunits of tetrameric voltage-gated cation channels. Attempts to express CatSper1 functionally in heterologous expression systems, with or without the other CatSper subunits, have been unsuccessful, perhaps owing to a requirement for the intraflagellar transport system of sperm. Here we have directly recorded the activity of ion channels from the plasma membrane of intact mouse sperm cells in whole-cell and perforated-patch configurations. Using these techniques, we have recorded a Ca^{2+} current (that we call CatSper or I_{CatSper}) that originates specifically from the principal piece of the sperm cell flagellum and is absent in *Catsper*^{−/−} mice.

After numerous unsuccessful attempts to form a giga-ohm seal between the patch pipette and the plasma membrane of the sperm cell head, midpiece and tail, we surmised that the tight link between the plasma membrane and stiff intracellular structures prevented seal formation. We circumvented this problem by applying the patch pipette to the small cytoplasmic droplet of the sperm (Fig. 1a and Supplementary Movie), a remnant of the precursor germ cell cytoplasm located on the midpiece of a mature sperm cell that is shed around the time of ejaculation¹⁸. We measured from sperm in which the cytoplasmic droplet was adjacent to the principal piece, in order to record from the most mature sperm in the corpus epididymis¹⁸. We cannot rule out the possibility that the sperm acquire

additional properties with further transport through the male reproductive tract. The plasma membrane of the cytoplasmic droplet is loosely attached to the stiff structural elements of the sperm cell flagellum, allowing formation of a tight seal with the pipette and subsequent break-in to generate a stable whole-cell configuration. In this configuration, the pipette had electrical and diffusion access to all compartments of the sperm cell (head, middle and principal

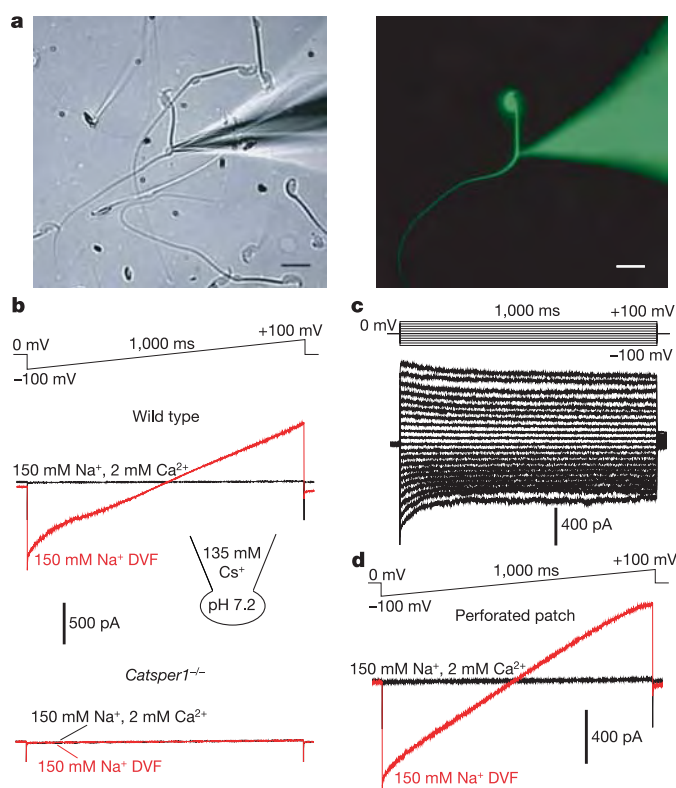


Figure 1 | Whole-cell patch-clamp recordings from intact spermatozoa. **a**, Left, the tight seal formed with the cytoplasmic droplet of the spermatozoon. Right, fluorescent image of the same spermatozoon after formation of the whole-cell configuration. Lucifer yellow (1 mM) diffused from the pipette throughout the interior of the spermatozoon in <5 s. Scale bar, 10 μm . **b**, Whole-cell currents elicited by a ramp from a holding potential (V_H) of 0 mV in DVF conditions from wild-type (top) and *Catsper1*^{−/−} (bottom) sperm. **c**, Representative I_{CatSper} in response to a step protocol from $V_H = 0$ mV in DVF conditions. **d**, I_{CatSper} recorded in perforated-patch mode of whole-cell recording.

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pieces; Fig. 1a). Mean capacitance of the whole sperm cell membrane was 2.5 ± 0.2 pF (consistent with $\sim 250 \mu\text{m}^2$ of sperm membrane; $n = 346$).

Because CatSper1 has been presumed to be a cationic channel⁴, the bath and pipette solutions contained minimal concentrations of membrane-permeable anions. Generally, K^+ ions were also omitted to prevent K^+ currents. Under such conditions and in the presence of 2 mM Ca^{2+} in the bath solution, the sperm cell membrane showed a very low conductance (Fig. 1b). On switching to a divalent-free (DVF) solution, however, we recorded a robust, quasilinear monovalent current, consistent with results from fluorescent dye measurements^{19,20}. The current was carried by Na^+ in the inward direction and Cs^+ in the outward direction (Fig. 1b, c). This current, which we call I_{CatSper} , was absent in *Catsper1*^{-/-} sperm cells (Fig. 1b, $n = 25$). We tested whether whole-cell mode perfusion of the cell interior with the pipette solution washed out key intracellular components that kept the channel in its closed state. In amphotericin perforated-patch recording, a method that prevents washout²¹, I_{CatSper} was unaltered as compared with the normal whole-cell configuration (Fig. 1d). Our results indicate that the CatSper channels are constitutively active and that CatSper1 is required for this current. Preliminary evidence suggests that the CatSper channel comprises the CatSper1 protein and other CatSper family subunits⁶. Complete identification of the channel constituents will require further investigation.

Another property of the CatSper1 protein is its specific localization to the principal piece of the sperm flagellum⁴, raising the issue of whether I_{CatSper} can be detected only in the principal piece. We recorded ionic currents from sperm cell fragments containing only the principal and midpiece, or only the head and midpiece (Fig. 2). I_{CatSper} was recorded from the fragment of the sperm cell containing the principal piece and midpiece (Fig. 2a), but not from the fraction consisting of only the head and midpiece (Fig. 2b). Therefore, I_{CatSper} originates from the principal piece of the sperm flagellum. The large amplitude of these currents (~ 500 pA/pF) suggests that CatSper channels densely localize to the flagellar plasma membrane.

The monovalent current through CatSper channels was potentially decreased by addition of Ca^{2+} to the extracellular side of the membrane (Fig. 3a), indicating a markedly high-affinity binding site for Ca^{2+} in the pore region of CatSper channels, which is a definitive property of Ca^{2+} -selective channels^{22,23}. We found that monovalent I_{CatSper} was strongly potentiated by intracellular alkalinization (Supplementary Fig. 1), and thus reasoned that we might detect the smaller, but physiologically relevant Ca^{2+} current by increasing the pH of the pipette solution. As bath $[\text{Ca}^{2+}]_o$ (that is, $[\text{Ca}^{2+}]_o$) was gradually raised from nominally 0 to 10 mM at $\text{pH}_{\text{pip}} 8.0$, inward Ca^{2+} current increased (Fig. 3b). A similar inward Ba^{2+} current

could also be elicited by a gradual increase in $[\text{Ba}^{2+}]_o$ from 0 to 50 mM (Fig. 3b). Inward current was not observed when the principal cation was Mg^{2+} (up to 50 mM $[\text{Mg}^{2+}]_o$; Fig. 3b). In recordings from *Catsper1*^{-/-} sperm cells, neither Ca^{2+} ($n = 12$) nor Ba^{2+} ($n = 14$) currents were detected under these conditions. We conclude that I_{CatSper} is a Ca^{2+} -selective current under physiological conditions. This CatSper Ca^{2+} current gradually inactivated at $[\text{Ca}^{2+}]_o > 2$ mM, but fully recovered after incubation in a nominal Ca^{2+} -free bath solution for several minutes (data not shown). To circumvent this rundown, $[\text{Ba}^{2+}]_o$ was used in subsequent experiments.

The extremely small free cytoplasmic volume of the flagellum and high density of active CatSper channels in the flagellar plasma membrane lead to significant shifts in intraflagellar concentrations of ions. This is seen as outward Ba^{2+} current in Fig. 3b, resulting from the accumulation of intraflagellar Ba^{2+} . Like inward Ba^{2+} current, outward Ba^{2+} current was absent in *Catsper1*^{-/-} mice. The channel activity is highest at depolarized potentials, as seen in Supplementary Fig. 2 where a strongly outwardly rectifying I_{CatSper} was recorded in symmetrical 50 mM Ba^{2+} . This current was not detected in *Catsper1*^{-/-} sperm cells. Under physiological conditions, I_{CatSper} should be active, voltage sensitive and highly Ca^{2+} selective.

I_{CatSper} shows weak voltage dependence, but the range of its voltage sensitivity is shifted significantly by intracellular alkalinization. Voltage-dependent activation of I_{CatSper} was measured from tail currents elicited after repolarization from various test potentials (Fig. 4a). The resulting activation curve shows that little current is active at pH 6.0 in the physiological range of potentials (about -70 to 0 mV). On alkalinization to pH 7.5, however, the activation curve is shifted to more hyperpolarized potentials. The potential at which half the channels are activated ($V_{1/2}$) shifts (-76 mV, from +87 mV at pH 6.0 to +11 mV at pH 7.5 (Fig. 4b)). The slope factor (k) of ~ 30 indicates weak voltage sensitivity as compared with slope factors of ~ 4 for strongly voltage-sensitive ion channels.

Indirect methods have suggested that Ca^{2+} fluxes are regulated by intracellular pH in sperm and spermatogenic cells^{24,25}. The intracellular pH of resting spermatozoa is acidic ($\text{pH} \approx 6.5$)^{26,27}. We recorded very small I_{CatSper} Ba^{2+} currents when the pipette pH was

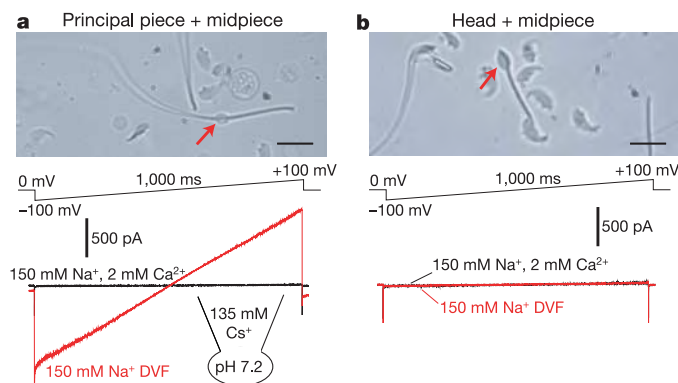


Figure 2 | I_{CatSper} originates from the principal piece of the sperm cell flagellum. **a**, Whole-cell recordings from a beheaded sperm cell containing only the mid- and principal piece (value $\pm$ s.e.m.: $C = 1.9 \pm 0.2$ pF; $n = 7$). **b**, Whole-cell recording from sperm cell head and midpiece ($C = 1.2 \pm 0.2$ pF; $n = 5$). Note the cytoplasmic droplet (arrow).

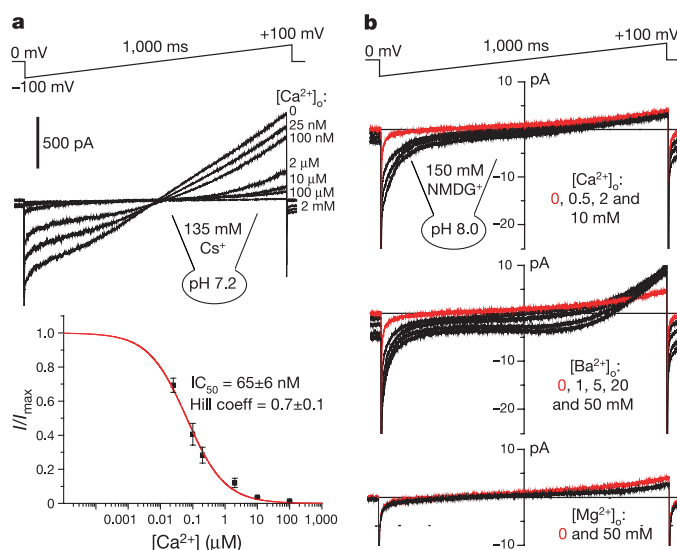


Figure 3 | I_{CatSper} is Ca^{2+} selective. **a**, Monovalent I_{CatSper} was potentially reduced in increasing $[\text{Ca}^{2+}]_o$. Top, representative I_{CatSper} traces elicited by a voltage ramp in the presence of different $[\text{Ca}^{2+}]_o$ ($V_H = 0$ mV). Bottom, relative I_{CatSper} amplitudes at -100 mV plotted against varying $[\text{Ca}^{2+}]_o$ (data are mean $\pm$ s.d.; $n = 8$). **b**, Divalent current elicited by a voltage ramp ($V_H = 0$ mV) with Ca^{2+} or Ba^{2+} as divalent ions. No inward current was detected in Mg^{2+} . '0' divalent concentration refers to nominal concentration.

6.5. As we increased the pH in the pipette, however, I_{CatSper} increased significantly (Fig. 4c). To buffer pH more accurately in the sperm cell, we replaced EGTA (which releases H^+ on binding Ba^{2+}) with BAPTA. Under these conditions and with reduced Ba^{2+} ($[\text{Ba}^{2+}]_o = 5 \text{ mM}$), I_{CatSper} increased from 2 ± 0.3 at $\text{pH}_{\text{pip}} 6.0$ to $12 \pm 1 \text{ pA}$ at $\text{pH}_{\text{pip}} 7.5$ ($n = 12$). I_{CatSper} could also be increased by intracellular alkalinization subsequent to bath application of $10 \text{ mM NH}_4\text{Cl}$ (Fig. 4d). Thus, capacitating conditions, which induce intracellular alkalinization in spermatozoa, should strongly potentiate I_{CatSper} . Notably, the amino terminus of CatSper1 protein is highly histidine rich (83 histidines in the 446-residue N terminus), suggesting that the N terminus has a role in sensing intracellular pH (ref. 4).

Cyclic nucleotides did not potentiate I_{CatSper} (Supplementary Fig. 3), suggesting that the observed CatSper1-dependent rise in intracellular Ca^{2+} (that is, $[\text{Ca}^{2+}]_i$) induced by cyclic nucleotides⁴ did not result from direct modulation of the CatSper channel. Calcium (2 mM) current through CatSper channels was not sensitive to the L-type Ca_V channels blockers nifedipine ($30 \mu\text{M}$) and nimodipine ($20 \mu\text{M}$). Cadmium ($200 \mu\text{M}$), nickel ($300 \mu\text{M}$) and ruthenium red ($10 \mu\text{M}$) reversibly blocked I_{CatSper} (data not shown).

In summary, we have provided a direct method for reliably reproducible, whole-cell recording of sperm cells. We have directly measured the principal Ca^{2+} -selective current in sperm, shown that it is greatly potentiated by intracellular alkalinization, and demonstrated that it is absent in *Catsper1*^{-/-} mice. To penetrate the cumulus and zona pellucida surrounding the egg, sperm hyperactivate, changing their tail motion from symmetric, fast and low amplitude (sinusoidal) to asymmetric, slow and large amplitude (whiplike)^{1,5}. An increase in $[\text{Ca}^{2+}]_i$ is required for hyperactivation²⁸; CatSper1 is required for mouse sperm hyperactivation⁵ and fertility⁴. Thus, we suggest that during sperm cell capacitation, alkalinization

activates the sperm-specific CatSper Ca^{2+} channel to increase intra-flagellar $[\text{Ca}^{2+}]$ and induce hyperactivated motility.

METHODS

Sperm cell preparation and incubation. Corpus epididymal sperm cells were prepared from male mice aged 3–8 months. Excised corpus epididymides were rinsed in HS medium comprising (in mM): 135 NaCl, 5 KCl, 2 CaCl_2 , 1 MgSO_4 , 20 HEPES, 5 glucose, 10 lactic acid and 1 sodium pyruvate; pH 7.4 (NaOH). Several incisions were made in the epididymis, and sperm cells were shaken from them in HS solution. After 10 min of settling time, the supernatant enriched with highly motile sperm cells was centrifuged, washed, resuspended in HS solution and stored at 25°C for up to 12 h. Beheaded and tail-less sperm cells were prepared by incubating the sperm cell suspension in the presence 0.2 mg ml^{-1} of trypsin at 37°C for 5 min, followed by gentle trituration.

Whole-cell recordings. The patch pipette was applied to the sperm cell cytoplasmic droplet. The cytoplasmic droplet migrates along the sperm cell midpiece from the head to the principal piece during spermatozoa epididymal transport and maturation. In most species the cell sheds it during ejaculation¹⁸. We used spermatozoa from corpus epididymis, because the cytoplasmic droplet of such cells was less fragile. The most mature spermatozoa, those with the cytoplasmic droplet located next to the principal piece, were chosen for experiments. Sperm were placed on untreated glass coverslips. Pipette resistance with caesium methanesulphonate solutions was 3–11 M Ω . Access resistance in the whole-cell configuration was 25–80 M Ω . Roughly five successful break-ins per day on average could be easily achieved. Cells were stimulated every 5 s.

The pipette solutions were made up as follows (in mM). For recordings of monovalent I_{CatSper} at intracellular pH (pH_i) 7.2: 135 caesium methanesulphonate, 5 CsCl, 10 HEPES, 10 EGTA, 5 Na_2ATP and 0.5 Na_2GTP (pH with CsOH); in some experiments, Na_2ATP and Na_2GTP were omitted from the pipette solution without any noticeable effect on I_{CatSper} . For recordings of monovalent I_{CatSper} at intracellular pH 6.0 and pH 8.4: 140 caesium methanesulphonate, 5 CsCl, 20 MES (pH 6.0) or TrisBase (pH 8.4), and 10 EGTA (pH with CsOH). For recordings of divalent currents at pH_i 7.0 and pH_i 8.0: 150 NMDG, 5 CsCl, 10 HEPES, 10 EGTA, 5 Na_2ATP and 0.5 Na_2GTP (pH with methanesulphonic acid); in some experiments, Na_2ATP and Na_2GTP or CsCl were omitted from the pipette solution without any noticeable effect on I_{CatSper} . For recordings of divalent currents at pH_i 6.0 and pH_i 6.5: 165 NMDG, 10 MES and 10 EGTA (pH with methanesulphonic acid). For recordings of divalent currents in symmetrical 50 mM Ba^{2+} conditions: 50 $\text{Ba}(\text{OH})_2$, 90 NMDG and 20 HEPES; pH 8.0 (with methanesulphonic acid). BAPTA solution (175 NMDG, 10 BAPTA, 10 HEPES and 10 MES; pH 6.0 or pH 7.5 with methanesulphonic acid) was used in some experiments in place of EGTA solutions (to eliminate divalent-dependent H^+ release by this chelator).

Seals between the patch pipette and the cytoplasmic droplet were formed in HS bath solution. After break-in, the bath solution could be changed as follows. For monovalent currents in divalent-free conditions (DVF solution): 150 sodium gluconate, 20 HEPES and 5 Na_3HEDTA ; pH 7.4 with NaOH. In some experiments, the DVF solution was modified by substitution of 5 Na_3HEDTA by 2 Na_3HEDTA and 2 EGTA, or by 1 Na_3HEDTA and 1 EGTA. CaCl_2 was added to the '2 Na_3HEDTA and 2 EGTA' DVF solution in accordance with the WinMAXC v2.05 program (C. Patton, Stanford Univ.) to obtain free $[\text{Ca}^{2+}]$ in the nanomolar and low-micromolar ranges. Bath solutions with different concentrations of Ca^{2+} , Ba^{2+} and Mg^{2+} ions were prepared by mixing nominal DVF bath solution (160 NMDG and 20 HEPES; pH 7.4 with methanesulphonic acid) and 50 mM divalent solution ($50 \text{ mM Ca}(\text{OH})_2$, $\text{Ba}(\text{OH})_2$ or $\text{Mg}(\text{OH})_2$, 90 NMDG and 20 HEPES; pH 7.4 with methanesulphonic acid). Osmolarity of the solutions was about 305 mmol kg^{-1} .

Perforated-patch recordings. Pipettes were filled as follows (in mM): 130 caesium methanesulphonate, 8 NaCl, 10 HEPES, 10 Cs_4BAPTA and $240 \mu\text{g ml}^{-1}$ of amphotericin B; pH 7.2. Pipette resistance was 12–17 M Ω . After antibiotic partitioning, access resistance in the whole-cell configuration was 35–100 M Ω . Bath solutions were similar to those used for the conventional whole-cell configuration.

Data analysis. Signals were acquired in PClamp 9 (Axon Instruments). Signals were filtered at 2 kHz and sampled at 10 kHz. The Hill equation was used to fit Fig. 3a, and the Boltzmann equation to fit Fig. 4b (Origin 7.0). Statistical data were calculated as the mean $\pm$ s.e.m.

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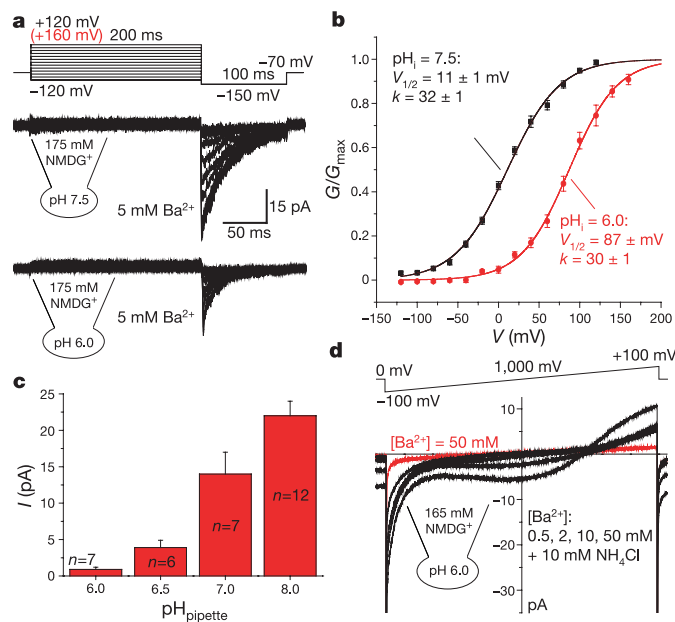


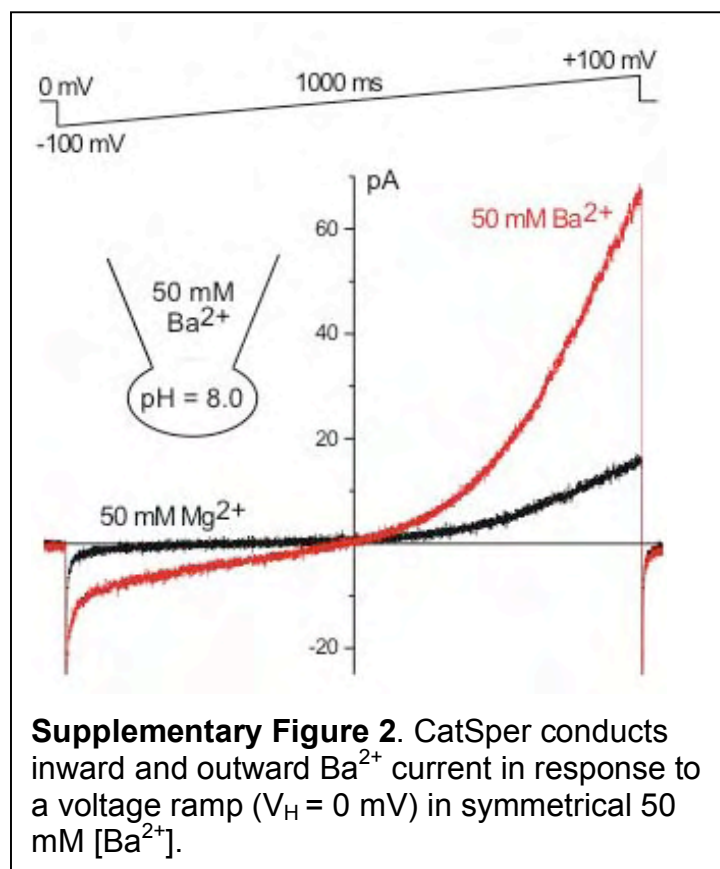
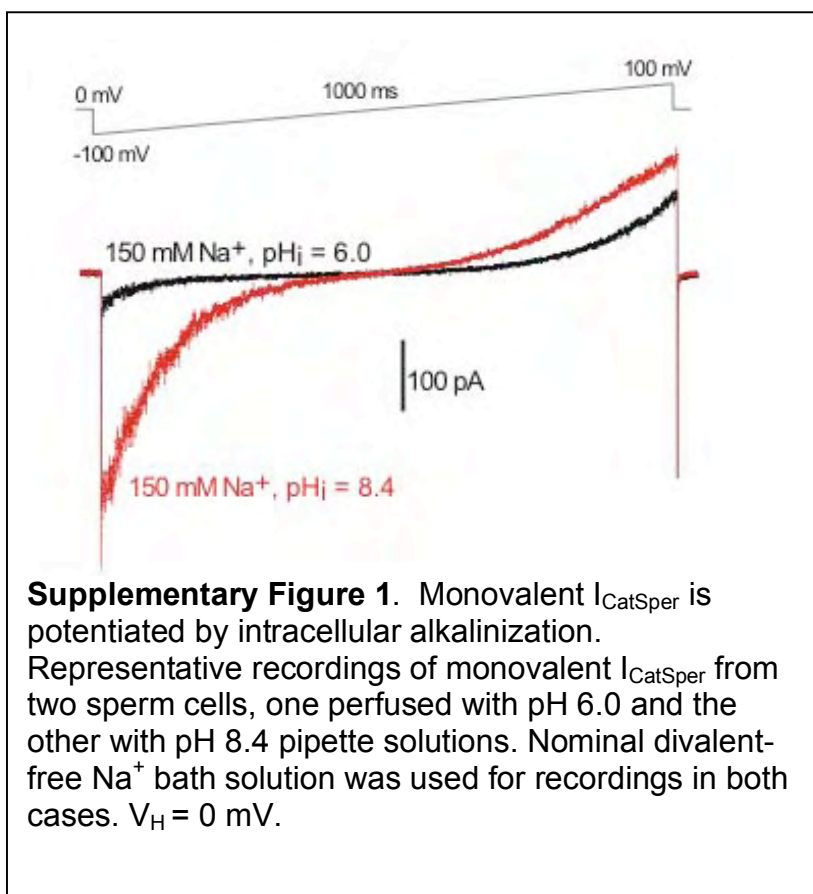
Figure 4 | pH and voltage-dependence of I_{CatSper} . **a**, I_{CatSper} relaxation (tail currents after subtraction of current in $5 \text{ mM } [\text{Mg}^{2+}]_o$) after instantaneous jump to -150 mV from voltages ranging from -120 mV to $+160 \text{ mV}$ ($V_H = -70 \text{ mV}$) at pH_i 7.5 (top) and pH_i 6.0 (bottom). **b**, Activation curves (fit to Boltzmann equation) of I_{CatSper} at pH_i 7.5 and pH_i 6.0, showing that alkalinization increases I_{CatSper} in the physiological range of potential (data are mean $\pm$ s.e.m.). **c**, Histogram showing the dependence of the mean I_{CatSper} Ba^{2+} amplitude on pH_{pip} . Ba^{2+} tail currents were elicited by stepping V_H from 0 to -100 mV . Bath $[\text{Ba}^{2+}] = 50 \text{ mM}$; current was measured at -100 mV after subtraction of the capacitive transients. **d**, Addition of $10 \text{ mM NH}_4\text{Cl}$ to the bath solution elicited robust Ba^{2+} I_{CatSper} ($\text{pH}_{\text{pip}} = 6$; $V_H = 0 \text{ mV}$).

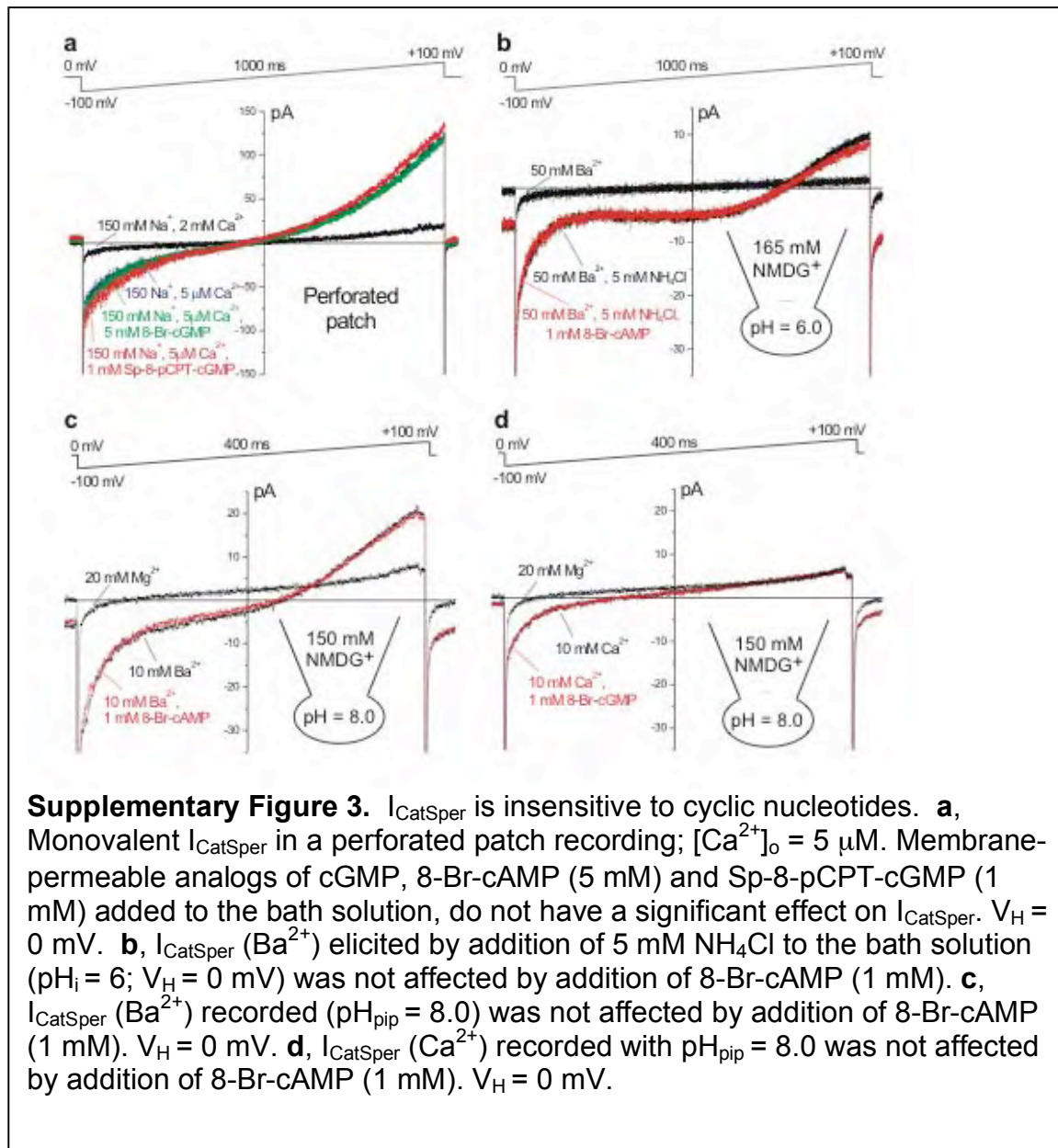
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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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Supplementary Figure 3. I_{CatSper} is insensitive to cyclic nucleotides. **a**, Monovalent I_{CatSper} in a perforated patch recording; $[\text{Ca}^{2+}]_o = 5 \mu\text{M}$. Membrane-permeable analogs of cGMP, 8-Br-cAMP (5 mM) and Sp-8-pCPT-cGMP (1 mM) added to the bath solution, do not have a significant effect on I_{CatSper} . $V_H = 0 \text{ mV}$. **b**, I_{CatSper} (Ba^{2+}) elicited by addition of 5 mM NH_4Cl to the bath solution ($\text{pH}_i = 6$; $V_H = 0 \text{ mV}$) was not affected by addition of 8-Br-cAMP (1 mM). **c**, I_{CatSper} (Ba^{2+}) recorded ($\text{pH}_{\text{pip}} = 8.0$) was not affected by addition of 8-Br-cAMP (1 mM). $V_H = 0 \text{ mV}$. **d**, I_{CatSper} (Ca^{2+}) recorded with $\text{pH}_{\text{pip}} = 8.0$ was not affected by addition of 8-Br-cAMP (1 mM). $V_H = 0 \text{ mV}$.